

Assignment 3

GitHub Project to EC2 Deployment using Jenkins Freestyle

Objective

Deploy a website from the GitHub repository Shantanumajan6/project to an AWS EC2 Linux VM using a Jenkins Freestyle project. Jenkins checks out the repository, installs and starts Apache HTTP Server if required, copies the project files to /var/www/html, and verifies that the website is accessible through the EC2 public IP.

Architecture / Flow

GitHub Repository → Jenkins Freestyle Job → Git Checkout → Jenkins Workspace → Install/Start Apache → Deploy Files → /var/www/html → Apache :80 → Browser

Item	Value
Source	GitHub repository
Repository	https://github.com/Shantanumajan6/project
Branch	main
Jenkins Job	Project-Freestyle-Deployment
Jenkins Type	Freestyle Project
Destination	AWS EC2 Linux VM
Web Server	Apache HTTP Server (httpd)
Document Root	/var/www/html
HTTP Port	80

Jenkins Freestyle Deployment Steps

1. Verify the Linux VM

Connect to the EC2 VM using SSH.

```
ssh -i "key_pair.pem" ec2-user@ec2-52-66-250-166.ap-south-1.compute.amazonaws.com
```

Note: Apache installation and service management will be performed by the Jenkins Freestyle build.

```
+ MobaXterm Personal Edition v26.3 +
(X server, SSH client and network tools)

► Your computer drives are accessible through the /drives path
► Your DISPLAY is set to 192.168.1.38:0.0
► When using SSH, your remote DISPLAY is automatically forwarded
► Each command status is specified by a special symbol (✓ or ✕)

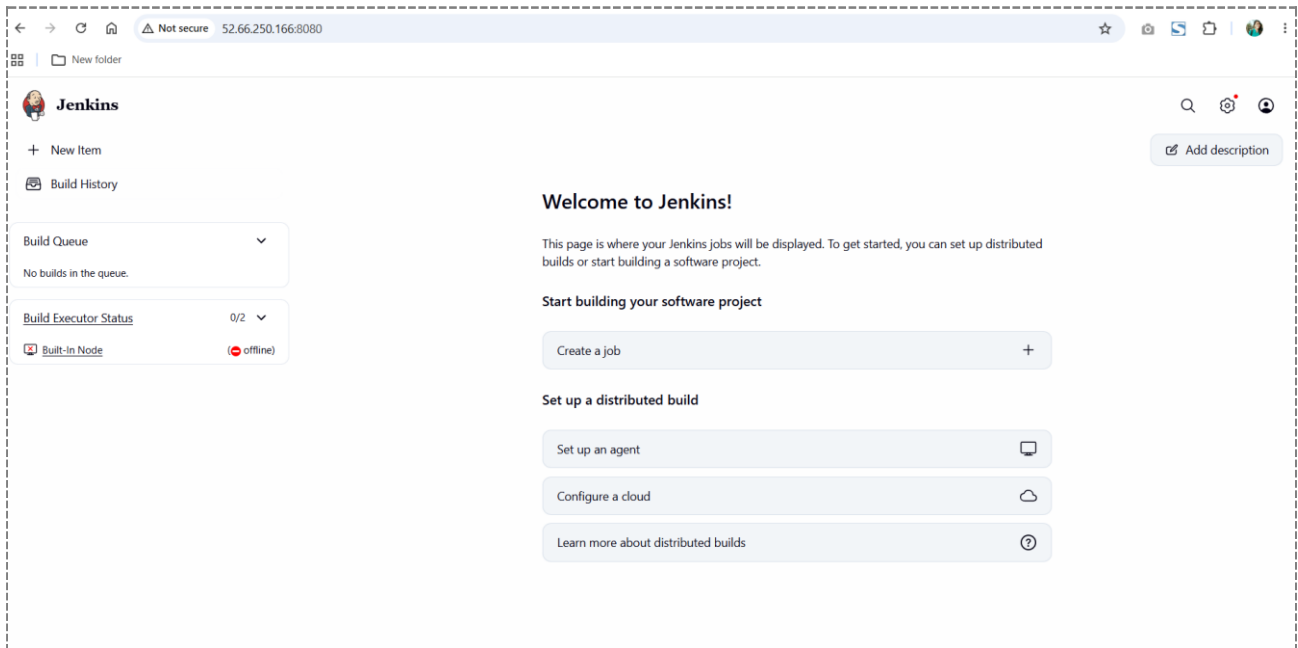
• Important:
This is MobaXterm Personal Edition. The Professional edition
allows you to customize MobaXterm for your company: you can add
your own logo, your parameters, your welcome message and generate
either an MSI installation package or a portable executable.
We can also modify MobaXterm or develop the plugins you need.
For more information: https://mobaxterm.mobatek.net/download.html

2026-10-03 11:43:06 /home/mobaxterm cd Desktop/aws_key/

2026-10-03 12:20:56 /home/mobaxterm/Desktop/aws_key ssh -i "key_pair.pem" ec2-user@ec2-52-66-250-166.ap-south-1.compute.amazonaws.com
X11 forwarding request failed on channel 0
#####
Amazon Linux 2023
#####
https://aws.amazon.com/linux/amazon-linux-2023
#####
Last login: Sat Oct 3 02:54:29 2026 from 103.115.201.70
[ec2-user@ip-172-31-1-51 ~]$ sudo su -
Last login: Sat Oct 3 02:55:27 UTC 2026 on pts/1
[root@ip-172-31-1-51 ~]#
```

2. Verify Jenkins

Verify Jenkins is running and open the Jenkins dashboard in the browser (<http://52.66.250.166:8080/>).



3. Install and Verify Git

```
yum install git -y
git --version
```

Git must be available on the Jenkins VM so that Jenkins can check out the GitHub repository.

```
[root@ip-172-31-1-51 ~]# git --version
git version 2.50.1
[root@ip-172-31-1-51 ~]#
```

4. Create Jenkins Freestyle Project

In Jenkins:

1. Click New Item
2. Enter: Project-Freestyle-Deployment
3. Select Freestyle project
4. Click OK

Jenkins / All / New Item

New Item

Enter an item name

Project-Freestyle-Deployment

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**
Creates a set of Pipeline projects according to detected branches in one SCM repository.
- Organization Folder**
Creates a set of multibranch project subfolders by scanning for repositories.
- Duplicate an existing item**
Create a new item by copying the configuration of an existing one.

OK

5. Configure Source Code Management

Select: Source Code Management → Git

Repository URL: <https://github.com/Shantanumajan6/project>

Branch: */main (use whichever branch holds the code, for example */master)

Note: Manual git clone is not required. Jenkins performs the Git checkout before executing the build steps.

6. Add Jenkins Execute Shell Build Step

Go to: Build Steps → Add build step → Execute shell

Use:

```
#!/bin/bash

echo "==== Starting Deployment ====="

echo "==== Jenkins Workspace ====="
pwd
ls -la

echo "==== Installing Apache ====="
sudo yum install httpd -y

echo "==== Starting Apache ====="
sudo systemctl enable httpd
sudo systemctl start httpd

echo "==== Deploying Website ====="
sudo cp -r ./ * /var/www/html/
sudo chown -R apache:apache /var/www/html
sudo chmod -R 755 /var/www/html

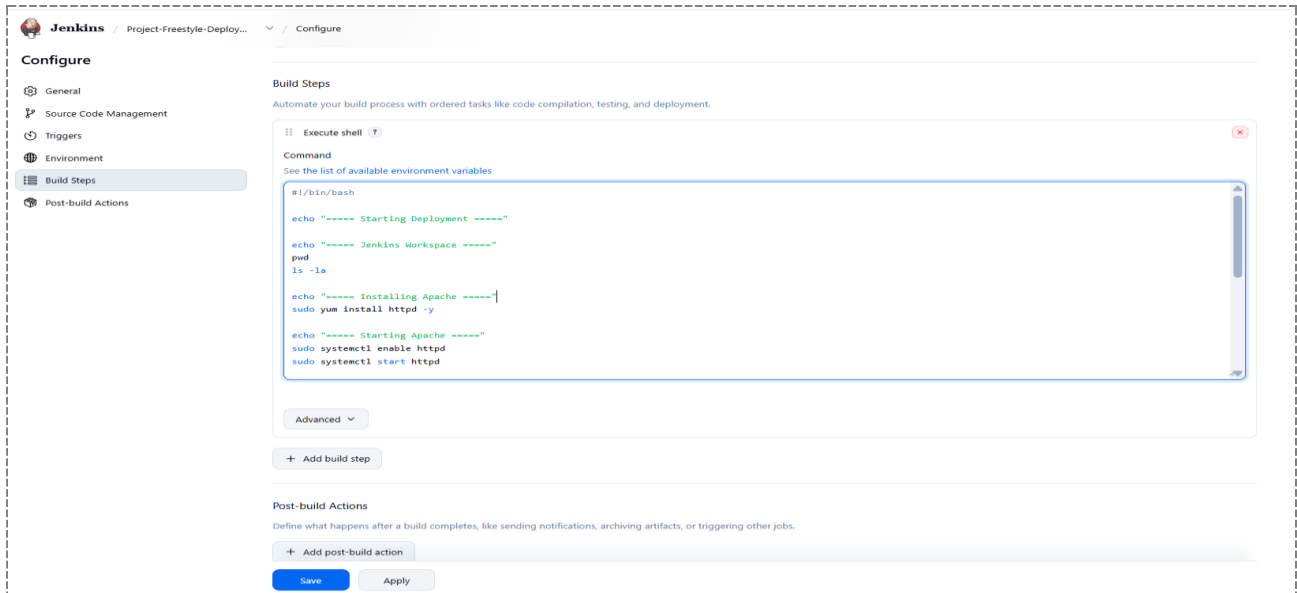
echo "==== Deployed Files ====="
```

```
ls -la /var/www/html/

echo "==== Apache Status ====="
sudo systemctl status httpd --no-pager

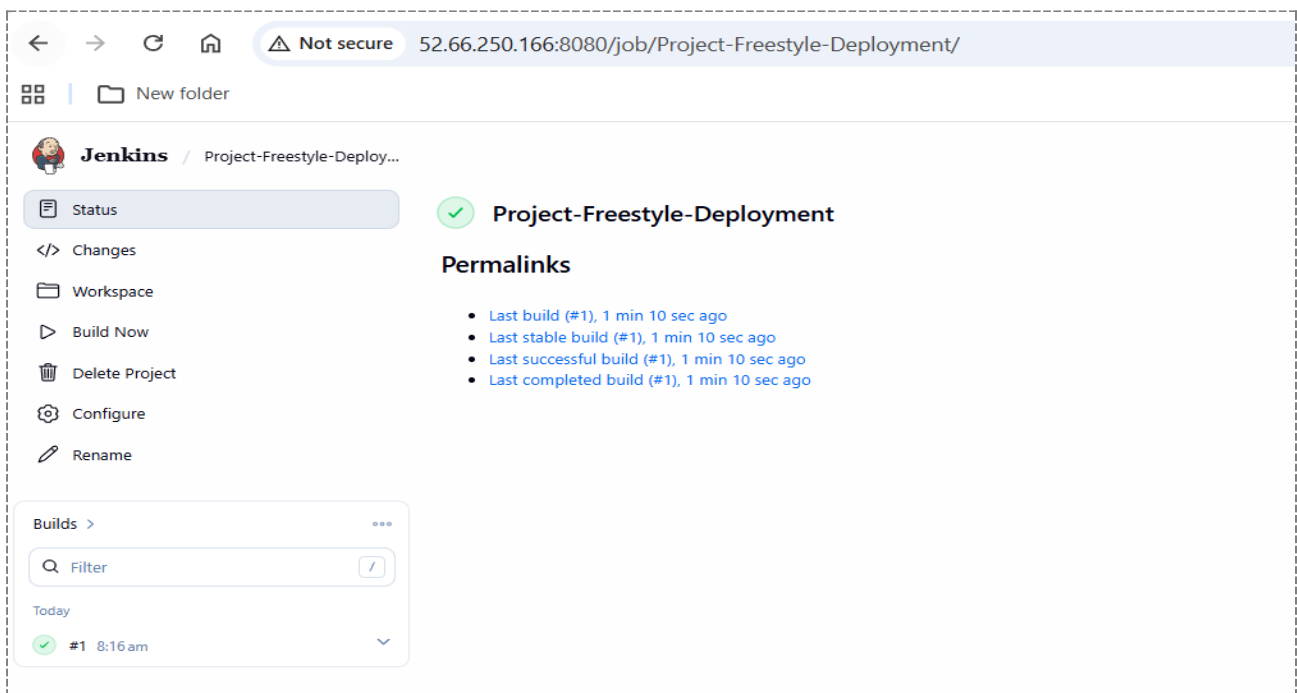
echo "==== Testing Website ====="
curl -I http://localhost

echo "==== Deployment Completed ====="
```



7. Save and Build

Click Save, then Build Now. Jenkins will check out the repository, install Apache, copy the files, and verify the deployment.



Open the build and select Console Output. Confirm the build ends with Finished: SUCCESS.

Jenkins

Project-Freestyle-Deploy...

#1

Status

Changes

Console Output

Delete build #1

Git Build Data

Edit build information

Download

Copy

View as plain text

Console

```

Started by user Adelin
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Project-Freestyle-Deployment
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/Shantanujana6/project
> git init /var/lib/jenkins/workspace/Project-Freestyle-Deployment # timeout=10
Fetching upstream changes from https://github.com/Shantanujana6/project
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/Shantanujana6/project +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/Shantanujana6/project # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision 4e0e5cf4feb2e7344c3d8059461524d8b9eb69 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f 4e0e5cf4feb2e7344c3d8059461524d8b9eb69 # timeout=10
Commit message: "Update web.xml"
First time build. Skipping changelog.
[Project-Freestyle-Deployment] $ /bin/bash /tmp/jenkins10735130050513243604.sh
===== Starting Deployment =====
===== Jenkins Workspace =====
/var/lib/jenkins/workspace/Project-Freestyle-Deployment
total 32
drwxr-xr-x. 6 jenkins jenkins 120 Oct 3 08:16 .
drwxr-xr-x. 10 jenkins jenkins 16384 Oct 3 08:16 ..
-rw-r--r--. 1 jenkins jenkins 1023 Oct 3 08:16 .classpath
drwxr-xr-x. 7 jenkins jenkins 146 Oct 3 08:16 .git
drwxr-xr-x. 3 jenkins jenkins 23 Oct 3 08:16 .github
-rw-r--r--. 1 jenkins jenkins 861 Oct 3 08:16 .project
drwxr-xr-x. 2 jenkins jenkins 133 Oct 3 08:16 .settings
-rw-r--r--. 1 jenkins jenkins 507 Oct 3 08:16 Jenkinsfile
-rw-r--r--. 1 jenkins jenkins 949 Oct 3 08:16 pom.xml
drwxr-xr-x. 3 jenkins jenkins 18 Oct 3 08:16 src
===== Installing Apache =====
Last metadata expiration check: 5:10:46 ago on Sat Oct 3 03:05:55 2026.
Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
===== Starting Apache =====
===== Deploying Website =====
===== Deployed Files =====
total 16
drwxr-xr-x. 6 apache apache 115 Oct 3 08:16 .
drwxr-xr-x. 4 root root 33 Oct 3 03:12 ..
-rwxr-xr-x. 1 apache apache 507 Oct 3 08:16 Jenkinsfile
drwxr-xr-x. 2 apache apache 24 Oct 3 06:19 R1
drwxr-xr-x. 2 apache apache 24 Oct 3 06:19 R2
drwxr-xr-x. 2 apache apache 24 Oct 3 06:19 R3
-rwxr-xr-x. 1 apache apache 27 Oct 3 07:23 dev.html
-rwxr-xr-x. 1 apache apache 30 Oct 3 07:23 index.html
-rwxr-xr-x. 1 apache apache 949 Oct 3 08:16 pom.xml
drwxr-xr-x. 3 apache apache 18 Oct 3 08:16 src
===== Apache Status =====
* httpd.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
Active: active (running) since Sat 2026-10-03 03:14:37 UTC; 5h 2min ago
Docs: man:httpd.service(8)
Main PID: 28835 (httpd)
Status: "Total requests: 38; Idle/Busy workers 100/0; Requests/sec: 0.0021; Bytes served/sec: 2 B/sec"
Tasks: 230 (limit: 1059)
Memory: 18.8M
CPU: 20.591s
CGroup: /system.slice/httpd.service
└─28835 /usr/sbin/httpd -DFOREGROUND
└─28858 /usr/sbin/httpd -DFOREGROUND
└─28860 /usr/sbin/httpd -DFOREGROUND
└─28861 /usr/sbin/httpd -DFOREGROUND
└─28869 /usr/sbin/httpd -DFOREGROUND
└─29135 /usr/sbin/httpd -DFOREGROUND

Oct 03 03:14:37 ip-172-31-1-51.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Oct 03 03:14:37 ip-172-31-1-51.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Oct 03 03:14:37 ip-172-31-1-51.ap-south-1.compute.internal httpd[28835]: Server configured, listening on: port 80
===== Testing Website =====
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left   Speed

  0   0   0    0   0    0     0      0      0      0   0      0   0
  0  30   0    0   0    0     0      0      0      0   0      0   0
  0 100   0    0   0    0     0      0      0      0   0      0   0
  0 100   0    0   0    0     0      0      0      0   0      0   0

HTTP/1.1 200 OK
Date: Sat, 03 Oct 2026 08:16:43 GMT
Server: Apache/2.4.68 (Amazon Linux)
Last-Modified: Sat, 03 Oct 2026 07:23:20 GMT
ETag: "1e-65cea5f87af7"
Accept-Ranges: bytes
Content-Length: 30
Content-Type: text/html; charset=UTF-8

===== Deployment Completed =====
Finished: SUCCESS

```

Jenkins 2.580.1

One Jenkins build demonstrates the full flow: Git checkout → Apache installation → Apache startup → website deployment → permissions → verification.

8. Verify Deployment on EC2 VM

After a successful Jenkins build, SSH into the VM and run:

```
ls -la /var/www/html/
```

Expected: the project files from the repository (for example index.html).

Then:

```
curl -I http://localhost
```

Expected: HTTP/1.1 200 OK

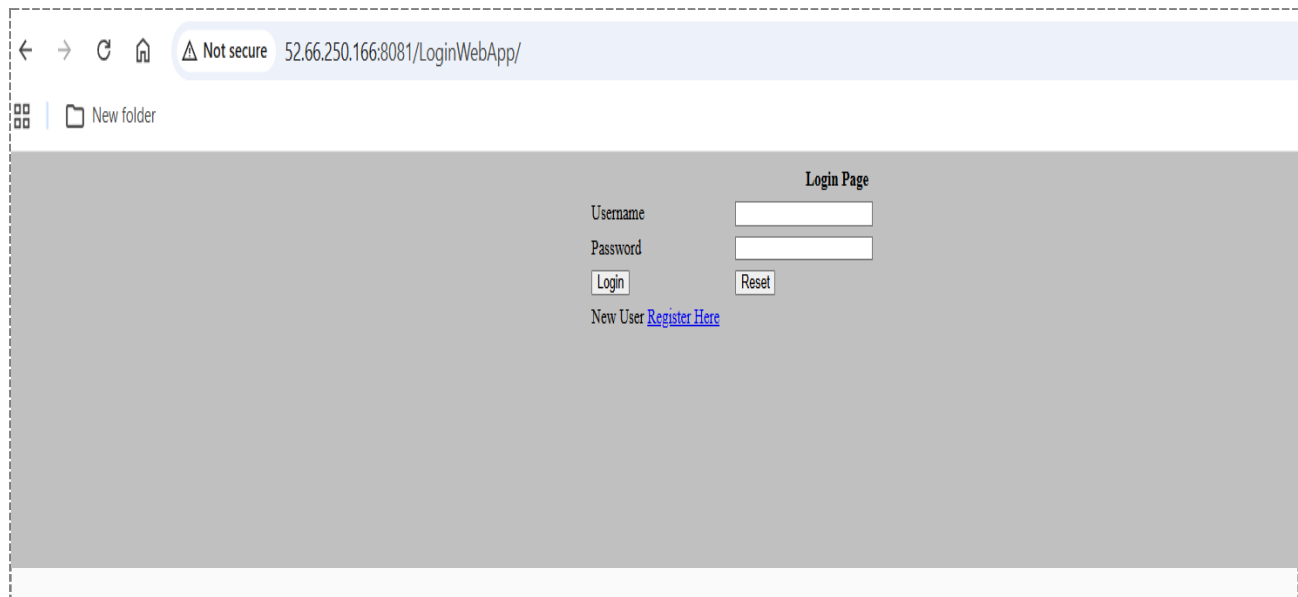
```
[root@ip-172-31-1-51 html]# curl -I http://localhost
HTTP/1.1 200 OK
Date: Sat, 03 Oct 2026 08:21:31 GMT
Server: Apache/2.4.68 (Amazon Linux)
Last-Modified: Sat, 03 Oct 2026 07:23:20 GMT
ETag: "1e-65cea85f87af7"
Accept-Ranges: bytes
Content-Length: 30
Content-Type: text/html; charset=UTF-8

[root@ip-172-31-1-51 html]#
```

9. Verify Website from Browser

Open: <http://52.66.250.166:8081/LoginWebApp/>

Note: Make sure port 81 is open in the EC2 Security Group (inbound rule: HTTP, TCP 81).



Outcome

- Freestyle job pulls code from the GitHub repository.
- Apache is installed and started automatically.
- Website files are deployed to /var/www/html.
- The website is accessible via the EC2 public IP.